Modeling single-beam sonar with Gazebo ray tracing[1]

1 Introduction

The complexity of the sea poses difficulties in effective communication and navigation which makes it difficult to achieve full autonomy in UUVs.

The risk that unreliable communications or navigation posed by unmanned vehicles continues to be an issue.

In-situ or field testing is required to validate a new technology, control algorithm, or proof-of-concept.

Difficult environments and other factors can sometimes cause delays in testing, creating long periods of time between design and testing.

Simulation environments are useful tools for testing autonomous vehicles.

They provide a cost-effective capability to provide rapid feedback on system updates, proof-of-concepts, etc.

Simulation is not a substitute for field testing.

Capturing the physics of the underwater environment is computationally expensive.

There are few simulators capable of simulating an underwater environment and fewer underwater sensors.

The motivation for this paper si to provide the underwater development community a sonar sensor model that can be used in research and development efforts for UUVs.

We present a computationally efficient sonar sensor model and implementation that generates data consistent with experimental measurements typical of high-frequency forward-looking (FLS) and multibeam sonar (MBS) sensors.

The sonar sensor model includes several acoustic properties to capture underwater physical phenomena such as target strength (TS), transmission loss (TL), speckle noise, and scattering. These properties are captured in a point-scattering method developed in [1].

Both point-scattering and beamforming are taken into account, making our implementation unique among sonar sensor models.

2 Related Work

Work has been published describing the development of a sonar simulation model.

Two of these have developed their models into supporting robotics framework.

In [2] a Gazebo sonar sensor model is developed using ray tracing. The Gazebo ray-tracing functionality generates a 3D point cloud which is transformed into a sonar image. On inspection, the acoustic properties were either hard-coded or commented out and did not include speckle noise simulation.

[1] Document source at `https://www.overleaf.com/read/tcxjgnmqphwj` and `https://github.com/bsb808/OceanNotes`

Model	DeMarco et al. (2015)	Cerquiera et al. (2017)	Ours
Sonar Type	FLS	FLS MSIS	FLS MB
Rendering	Rasterization	Rasterization	Rasterization
Features	Reflection Speckle	Reflection Speckle Attenuation Surface Irregularities Reverberation	Reflection Speckle Attenuation Scattering Beamforming
Evaluation	Qualitative Computation Time	Qualitative Quantitative	Qualitative Compuation Time

Table 1: Summary of Sonar Sensor Models

In [3], a GPU-based sonar simulator is developed using rasterization. They model two types of sonar: mechanically scanned imaging sonar (MSIS) and FLS. The acoustics features provided in their model are accurate and representative of sound propagation. The sonar sensor model is modeled using a virtual camera and utilizes the following three parameters to render the camera as sonar: pulse distance, echo intensity, and field-of-view. Physical tests were also conducted to compare the simulation results with the physical imaging sonar.

3 Approach

The model is based on a ray-based spatial descretization of the model facets, beampattern appropriate for a line array, and a point scattering model of the echo level.

3.1 Active Sonar Equation

With respect to a target of interest, the simplest form of the sonar equation is given as

$$EL = SL - 2(TL) + TS \tag{1}$$

where echo level, EL, is the intensity of the returned signal which is a function of the source level, SL, two-way transmission loss, TL, and target strength, TS.

SL is the sound pressure level at a distance of 1m from a sonar and is defined as the decibel of the ratio of the signal intensity, I_0, and a reference intensity, I_{ref}. A common I_{ref} in underwater acoustics is a reference pressure of $1\,\mu\text{Pa}$. SL is defined as

$$SL = 10 \log \frac{I_0}{I_{ref}} \tag{2}$$

TL is the attenuation of sound as it propagates a distance $1\,\text{m}$ from a sonar source to a range, r, from a source.

$$TL = 20 \log \frac{r}{r_0} \tag{3}$$

where r_0 is a reference range typically given as $1\,\mathrm{m}$. Attenuation loss can be due to absorption, scattering, and leakage from sound channels [4]. Absorption loss can be modeled by the absorption coefficient, a_c, as

$$a_c = \frac{A_1 P_1 f_1 f^2}{f_1^2 + f^2} + \frac{A_2 P_2 f_2 f^2}{f_1^2 + f^2} + A_3 P_3 f^2 \tag{4}$$

where f is the transmitted frequency, f_1 and f_2 are the boric acid and magnesium sulfate relaxation frequencies, A_1, A_2, and A_3 are weighting amplitudes, P_1, P_2, and P_3 are nondimensional pressure correction factors [5]. The attenuation coefficient is calculated by

$$\alpha_p = \frac{a_c \log 10}{20} \tag{5}$$

The wave vector k_w can be related to the attenuation via the overall attenuation which is modeled using a complex frequency-dependent relationship between the wave vector and the attenuation coefficient

$$K = k_w + i\alpha_p \tag{6}$$

where k_w is defined as

$$k_w = \frac{2\pi f}{c} \tag{7}$$

According to [4], the combination of spherical spreading and absorption provides a reasonable approximation under a wide variety of conditions,

$$TL = 20 \log(r) + a_c r \tag{8}$$

Target strength is defined as the ratio of intensity reflected, I_r by an object at a distance of $1\,\mathrm{m}$ from the object to the incident intensity, I_i,

$$TS = 10 \log \frac{I_r}{I_i} \tag{9}$$

True target strength in water depends on a variety of parameters such as specular reflection, object aspect, surface irregularities, acoustic impedance, and internal structure and composition.

3.2 Single Beam Sonar Model

A single sonar beam is modeled using discrete rays. The individual rays are indexed as $i = \{1, 2, ...N\}$ for N rays. The following information is generated for each ray within an individual beam:

- The range r_i as the distance from the origin of the sonar reference frame to the first intersection between the ray and a target in the field of view. The azimuth of the ray is fixed in the sensor frame as θ_i and the inclination angle of the ray is ϕ_i

- The incident angle a_i as the difference between the ray vector, $\mathbf{z}$, and the normal vector, $\mathbf{n}$ of the target surface at the location of intersection between the ray vector and the target surface.
- The reflectivity of the target intersected by the i^{th} ray, $\mu_i cos(\theta)$, which is a property of the target object model

3.2.1 Ray-based Beam Model

We define a ray as a vector, $\mathbf{z}$, from the sensor frame origin to the first intersection with a visual object within the scene. We calculate the incident angle, α_i, as

$$\alpha_i = 180^\circ - cos^{-1}(\mathbf{\hat{z}_i} \cdot \mathbf{\hat{n}_i}) \tag{10}$$

where $\mathbf{\hat{z}_i}$ and $\mathbf{\hat{n}_i}$ are the ray and normal direction unit vectors respectively.

The projected ray surface area, dA, is the area projected onto the visual object by the individual ray. If the changes in both $d\theta_i$ and $d\phi_i$ angles for each ray are assumed to be infinitesimally small, then the projected area ray scene can be calculated by

$$dA = r_i^2 d\theta_i d\phi_i \tag{11}$$

The ray area projected onto a surface, dS_i is a function of the incident angle, calculated as

$$dS_i = \frac{dA}{cos(\alpha_i)} \tag{12}$$

and expressed as

$$dS_i = \frac{r_i^2 d\theta_i d\phi_i}{cos(\alpha_i)} \tag{13}$$

The target strength is given by

$$TS_i = 10 \log \frac{I_{ri}}{I_{Ii}} \tag{14}$$

and using Lambert's reflection law, the ratio of intensities is

$$\frac{I_{ri}}{I_{Ii}} = \mu cos^2(\alpha_i) dS_i \tag{15}$$

Substituting for dS_i, the intensity ratio becomes

$$\frac{I_{ri}}{I_{Ii}} = \mu cos(\alpha_i) r_i^2 d\theta_i d\phi_i \tag{16}$$

The physical model defined by Equation 6 is simulated using a point-based scattering model developed in [1]. The model generates a spatially coherent time series that is useful in simulating narrowband sonar applications such as the FLS and MBS systems. The model uses discrete scatterers distributed over a surface. These scatterers are representative of the number of surfaces a ray intersects based on the object's surface mesh. In our model, this approach is computationally inefficient. Therefore, we define each ray in a beam as a scatterer. The overall spectrum of the signal received from each ray, $P(f)$, is computed as

$$P(f)_i = \frac{S(f) \sum_{i=1}^{N} a_i e^{iKr_i}}{r_i^2} \tag{17}$$

where $S(f)$ is the transmitted spectrum of the source, N is the number of scatters, a_i is the complex scatterer amplitude, and f is a frequency vector. $P(F)$ is a combination of the physical model for echo level and a complex random scale factor for speckle noise resolved in the frequency domain.

The source level is a user defined input and remains constant for each ray and is modeled in the frequency domain by $S(f)$,

$$S(f)_i = S_0 e^{-(f-f_c)^2 b^2 \pi^2} \tag{18}$$

The frequency vector, f, is a linearly spaced vector from f_{min} to f_{max} and centered on the central frequency, f_c. The full width of the transmit spectrum is the bandwidth, b, a user provided input based on the sonar specifications. As an example, the bandwidth for the BlueView P900-45 FLS is $2.95\,\mathrm{kHz}$

$$f = f_{min} \cdots f_{max} \tag{19}$$

$$f_{min} = f_c - \frac{b}{2} \tag{20}$$

$$f_{max} = f_c + \frac{b}{2} \tag{21}$$

Spherical spreading is an appropriate assumption for modeling transmission loss. The two-way transmission loss for incoherent scattering in $P(f)$ is captured in the denominator, r_i^2

The scatter amplitude, a_i, is calculated for each ray,

$$a_i = \frac{\xi_{xi} + i\xi_{yi}}{\sqrt{2}} \sqrt{\mu cos^2(\alpha_i) r_i^2 d\theta_i d\phi_i} \tag{22}$$

Although the random variable, ξ_i, is indexed by i to represent the ray index, the real random variable and the complex random variable must both be generated and different from each other, hence the x and y notation. Overall, the random variables are representative of Gaussian noise and for our purposes, satisfies the speckle noise requirement [1]. The variables under the square root represent the target strength of an incident ray on an object.

3.3 Beam Pattern Geometry

We combine the ray-based discretization of the single beam with a linear array beam pattern model. The beam pattern of an array is defined in polar coordinates where the acoustic intensity is the distance along the radial axis and the angle is relative to the transducer axis. The beam pattern is visualized as one main lobe in the center with smaller side lobes radiating away from the main axis. By inspection, the highest return will be along the main axis as the response decreases off-axis. Therefore, the echo level depends on the size and position of a target within a beam. The beam width, θ_{bw}, is marked at $-3\,\mathrm{dB}$ on the main lobe. We define half the beam width as

$$\theta_w = \frac{\theta_{bw}}{2} \tag{23}$$

The beam pattern, $B(\theta)$, expresses the EL as a function of θ, the angle relative to the main axis

$$EL = EL|B(\theta)|^2 \tag{24}$$

For a continuous line array of length L, radiating energy at a wavelength λ, the beam pattern is that of a uniform aperture function. The radiated power is modeled as a normalized $sinc$ function

$$|B(Lu)|^2 = |sinc(Lu)|^2 = \begin{cases} 1, & \text{for } Lu = 0 \\ \left|\frac{sin(\pi Lu)}{\pi Lu}\right|^2, & \text{otherwise} \end{cases} \tag{25}$$

where u is the electrical angle

$$u = \frac{sin(\theta)}{\lambda} \tag{26}$$

The half intensity point, θ_w, can be solved for by setting

$$|B(\theta_w)|^2 = \left|\frac{sin\left(\pi\frac{L}{\lambda}sin(\theta_w)\right)}{\pi\frac{L}{\lambda}sin(\theta_w)}\right|^2 = \frac{1}{2} \tag{27}$$

For high frequency sonar, we can assume $L \gg \lambda$, and θ_w becomes

$$\frac{L}{\lambda} = \frac{0.884}{\theta_{bw}} = \frac{0.442}{\theta_w} \tag{28}$$

The final beam pattern is

$$|B(\theta_w)|^2 = \left|\frac{sin\left(\pi\frac{0.884}{\theta_{bw}}sin(\theta)\right)}{\pi\frac{0.884}{\theta_{bw}}sin(\theta)}\right|^2 \tag{29}$$

The azimuth angle, θ, is defined as rotation about the z sensor frame axis and the main axis is defined as coincident with the sensor frame z axis. The use of N rays, with angular location distributed evenly between in the range $[-\theta_{min}, \theta_{max}]$ and the intensity is a function of θ for the main lobe only.

The radial dimension is descretized by considering M samples within the interval $[R_{min}, R_{max}]$. The number of samples in the radial dimension is a function of the range interval and the range resolution (dR) and can be expressed as

$$M = \left| \frac{R_{max} - R_{min}}{dR} \right| \tag{30}$$

As an example, the BlueView P900-45 FLS has a specified range interval of 2-60 m with a range resolution of $dR = 0.0254\,\mathrm{m}$ resulting in $M = 2283$ samples per beam.

The single sonar beam model generates a single pair of range and intensity values, (r, I), which is generated by taking a uniform sample of the beam pattern. Each sample is a ray at θ_k in the range $[-\theta_w, theta_w]$ with the associated range and intensity pair (r_k, i_k). The set of ordered pairs from all rays is $R = \{(r_0, i_0), (r_1, i_1), ..., (r_N, i_N)\}$.

Using a uniform weighted average to model the beam pattern effect, the echo level for the beam can be calculated as

$$\overline{EL} = \bar{l} = \frac{\sum_{k=0}^{N} w_k i_k}{\sum_{k=0}^{N} w_k} \tag{31}$$

The weights are determined from the beam pattern function and calculated as

$$w_k = |B(\theta_k)|^2 \tag{32}$$

Algorithm 1 Sonar Model

```
ray ← calculateRay(distance, α)
S(f) ← calculateTransmitSpectrum(frequency)
for k = 1 : nBeams do
    for i = 1 : nRays do
        noise ← Gaussian(x, y)
        amplitude ← calculateAmplitude(noise, α, distance)
        P(f) ← calculateReceiveSpectrum(S(f), amplitude, distance)
    end for
    beampattern ← calculateBeamPattern(θ)
    P(t) ← transformReceiveSpectrum
    P(t)modified ← calculateModifiedReceiveIntensity(P(t), beampattern)
end for
SPL ← calculateSoundPressureLevel(P(t)modified)
```

References

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